What social media told us in the time of COVID-19: a scoping review

Automated LLM-assisted Review Pipeline

Abstract

This scoping review synthesizes 193 studies to understand the roles that social media played during the COVID-19 pandemic. From an initial pool of 6,253 records, a structured screening and evidence-card extraction process identified eight key themes, to which studies could belong simultaneously. Across themes, social media emerged as a double-edged tool: it facilitated real-time disease surveillance and public health communication, but also accelerated the spread of misinformation and stigmatizing narratives. For instance, social media data predicted case counts days ahead of official reports Shen et al. [2020], Hswen et al. [2020], while false claims propagated faster than accurate information Islam et al. [2020b]. Public sentiment analyses consistently documented heightened anxiety and depression Ahmad and Murad [2020], Radwan et al. [2020], although one study found no such association, suggesting moderating factors. Government communication often suffered from low engagement initially, with empathic and dialogic content proving more effective Liao et al. [2020]. Contrasts emerged across platforms: Weibo discussions focused on health, whereas Twitter emphasized policy and politics Chen et al. [2020b]. Methodological trends show increasing use of computational techniques such as natural language processing and network analysis. However, most research is limited to single platforms, short time frames, or English-language data, constraining generalizability. The review highlights the need for cross-platform, longitudinal, and culturally diverse approaches to capture the full scope of social media's influence during health crises.

1 Introduction

1.1 Background

The coronavirus disease 2019 (COVID-19) pandemic, caused by severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2), precipitated a global health crisis of unprecedented scale. As the virus spread rapidly across borders, public health authorities, researchers, and governments turned to digital platforms to disseminate information, monitor population attitudes, and coordinate response efforts. Social media platforms—including Twitter, Facebook, Sina Weibo, YouTube, and others—emerged as central arenas for public discourse, information exchange, and behavioural observation during the pandemic. These platforms generated vast quantities of user-generated content that captured real-time public reactions, concerns, and behaviours in the face of an evolving emergency.

Early in the pandemic, researchers recognised that social media could serve not only as a communication channel but also as a rich data source for understanding societal responses. Studies analysed tweets, posts, comments, and videos to track the spread of misinformation (the so-called "infodemic"), gauge public sentiment toward containment measures, detect emerging health concerns, and even predict case trends. At the same time, social media became a vector for conspiracy theories and unverified claims, complicating public health messaging and raising questions about platform governance. The dual role of social media—as both a sensor of public opinion and

a driver of information dynamics—prompted a surge of empirical investigations across disciplines such as epidemiology, computational social science, psychology, and communication.

1.2 Motivation and gap

Despite the rapid proliferation of studies on social media and COVID-19, the resulting evidence base remained fragmented. Individual investigations tended to focus on a single platform, a specific geographic region, or a narrow research question—for example, sentiment toward lockdowns in one country or the prevalence of misinformation on Twitter. Few attempts were made to synthesise the multiple functions that social media served during the crisis, to compare findings across methodologies, or to identify overarching themes in how social media shaped or reflected the pandemic experience. Moreover, the variety of data sources (e.g., Twitter, Weibo, Facebook, YouTube), analytical techniques (sentiment analysis, topic modelling, network analysis, machine learning), and outcome measures made it difficult to assess the cumulative evidence. A scoping review was therefore needed to map the breadth and nature of the existing literature, to clarify the roles that social media played, and to highlight evidence gaps that future research could address.

1.3 Review objective

The objective of this scoping review is to understand the roles that social media has had since the beginning of the COVID-19 crisis. Rather than assessing intervention effectiveness or quality, we aim to catalogue the types of research questions asked, the populations and platforms studied, the analytic approaches employed, and the principal findings reported in the first year of the pandemic.

1.4 Research questions

To guide the review, we formulated the following research questions:

1. What roles or functions have been attributed to social media in the context of the COVID-19 pandemic, from the onset of the crisis until November 2020?

2. Which platforms, geographic regions, and languages are represented in the peer-reviewed empirical literature, and how are they distributed?

3. What analytical methods and data types have been used to derive insights from social media data during this period?

4. What are the main thematic areas and evidence gaps in the current understanding of social media's role during the pandemic?

1.5 Contributions

This scoping review contributes a comprehensive map of the early academic response to the COVID-19 infodemic through social media analysis. By synthesising findings across studies that employ diverse data sources and methods, we identify common patterns, contrasts, and gaps. The review provides a structured overview of the multiple functions of social media—ranging from surveillance and sentiment monitoring to the spread and correction of misinformation, the delivery of health education, and the expression of public attitudes toward government responses. In doing so, it serves as a reference for researchers, public health practitioners, and platform policy-makers who seek to understand how social media can be leveraged or regulated in future health emergencies.

2 Methods

2.1 Review design and objective

This scoping review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR) guidelines. The objective was to understand the roles that social media has played since the beginning of the COVID-19 crisis, including its use for surveillance, information dissemination, public attitude tracking, and mental health support.

2.2 Search sources and keyword strategy

We searched eight electronic databases: MEDLINE, PubMed, Scopus, PsycINFO, OpenAlex, Crossref, Semantic Scholar, and Europe PMC. The search strategy combined terms related to COVID-19 (e.g., "COVID-19", "coronavirus", "SARS-CoV-2", "2019-nCoV") with social media platforms (e.g., "Twitter", "Facebook", "Sina Weibo", "YouTube"), related concepts (e.g., "infodemic", "misinformation", "sentiment analysis", "public attitude"), and study design filters (e.g., "empirical", "original", "data"). The search was restricted to articles published between November 1, 2019 and November 4, 2020, and to studies written in English or Mandarin.

2.3 Eligibility criteria

Studies were included if they were original empirical research (including peer-reviewed preprints) that discussed the use of social media for COVID-19 research. Eligible contributions had to involve platforms such as Twitter, Facebook, Weibo, or YouTube, and could address topics such as infodemics, public attitudes, mental health, detection, prediction, government responses, prevention education, or misinformation. We excluded reviews, opinion pieces, editorials, commentaries, book chapters, letters, systematic reviews, meta-analyses, and protocols. Non-peer-reviewed articles and studies written in languages other than English or Mandarin were also excluded.

2.4 Article retrieval, cleaning, and ranking

A total of 6,253 records were downloaded from the databases. After DOI-based metadata cleaning and deduplication, 4,759 unique records remained. These records were ranked by relevance using a machine learning classifier trained on a seed set of eligible studies. The top-ranked 1,000 records constituted the screening pool for further evaluation.

2.5 Full-text acquisition and LLM-assisted evidence-card extraction

Full texts were sought for the top-ranked screening pool. A target of 200 records (the top-ranked screening target) was selected for evidence-card generation. A large language model (LLM) was used to extract structured evidence cards from each of these records. Of the 200 evidence cards, 196 were generated from full-text versions; the remaining 4 were based on available abstract and metadata information. After a final eligibility determination, 5 records were excluded, 2 remained uncertain, and 193 were included in the synthesis.

2.6 Evidence-card completeness and synthesis preparation

We conducted an evidence-card completeness and extraction audit to ensure consistency and accuracy of the extracted data; no formal quality assessment scores were employed because the review focused on mapping the breadth of evidence rather than appraising methodological rigor.

The audit confirmed that the 193 included evidence cards captured the key study characteristics needed for thematic synthesis. The final dataset comprised these 193 studies, which were then analysed to identify patterns, contrasts, and evidence gaps across the literature.

3 Results

3.1 Study selection flow

The selection flow began with 6 253 records downloaded from the database search. After DOI-based metadata cleaning and deduplication, 4 759 records remained. These 4 759 records were ranked by abstract and title relevance, forming the full ranked set. From this ranked set, the top 1 000 records were selected as the screening pool. Within that pool, the top 200 records were designated for LLM-based screening and article-card generation. A total of 200 article cards were generated. After screening, 193 records were included for synthesis, corresponding to the included set. Five records were excluded and two were marked as uncertain; no decisions were missing.

3.2 Candidate metadata and ranking coverage

Of the 6 253 initially downloaded records, 4 759 passed DOI metadata cleaning, representing the set of records with sufficient metadata for ranking. These 4 759 records were ranked using a combined abstract and title scoring strategy. From that ranked list, the top 1 000 records constituted the screening pool. The top 200 records from this pool were forwarded for full LLM screening and card generation, thereby covering the most relevant candidates according to the ranking model.

3.3 Full-text availability

Among the 200 article cards generated, full-text documents were available for 196 records. The remaining four cards were prepared using abstract and metadata only, as no full text could be retrieved. All 200 cards consisted of a single text chunk (chunk count = 1), indicating that each document was processed as a contiguous unit.

3.4 LLM screening decisions

The LLM screening decisions for the top 200 records yielded the following outcome: 193 records were classified as include, 2 as uncertain, and 5 as exclude. No records had missing decisions. The 193 included records were subsequently used for synthesis.

3.5 Evidence-card completeness and extraction audit

No formal quality scores were assigned; therefore, completeness is reported in terms of evidence-card coverage. Of 200 generated cards, 193 were used in the final synthesis. Full text was available for 196 cards. All cards had a chunk count of 1. The completeness audit confirms that the extraction was performed on the intended set of included records, with full text informing the vast majority of evidence cards.

3.6 Canonical review themes

Eight canonical themes were identified across the included studies. Because studies may belong to multiple themes, the sum of theme counts exceeds the number of included studies (193). The theme distribution is as follows:

- Misinformation and the Infodemic: 75 studies
- Public Sentiment and Mental Health: 64 studies
- Health Communication and Preventive Behaviors: 38 studies
- Government Communication and Public Engagement: 37 studies
- Disease Surveillance and Prediction (Infoveillance): 27 studies
- Public Concerns and Risk Perception: 24 studies
- Cross-Platform and Cross-Cultural Dynamics: 20 studies
- Stigma and Conspiracy Theories: 19 studies

These eight themes capture the principal research foci represented in the scoping review literature.

3.7 Automatically generated review assets

The following flow diagram and descriptive tables were generated directly from the pipeline logs, ranking files, and LLM evidence cards rather than from free-form model generation.

Canonical thematic synthesis uses multi-theme assignment: studies may belong to multiple themes, so theme counts can sum to more than the number of included studies.

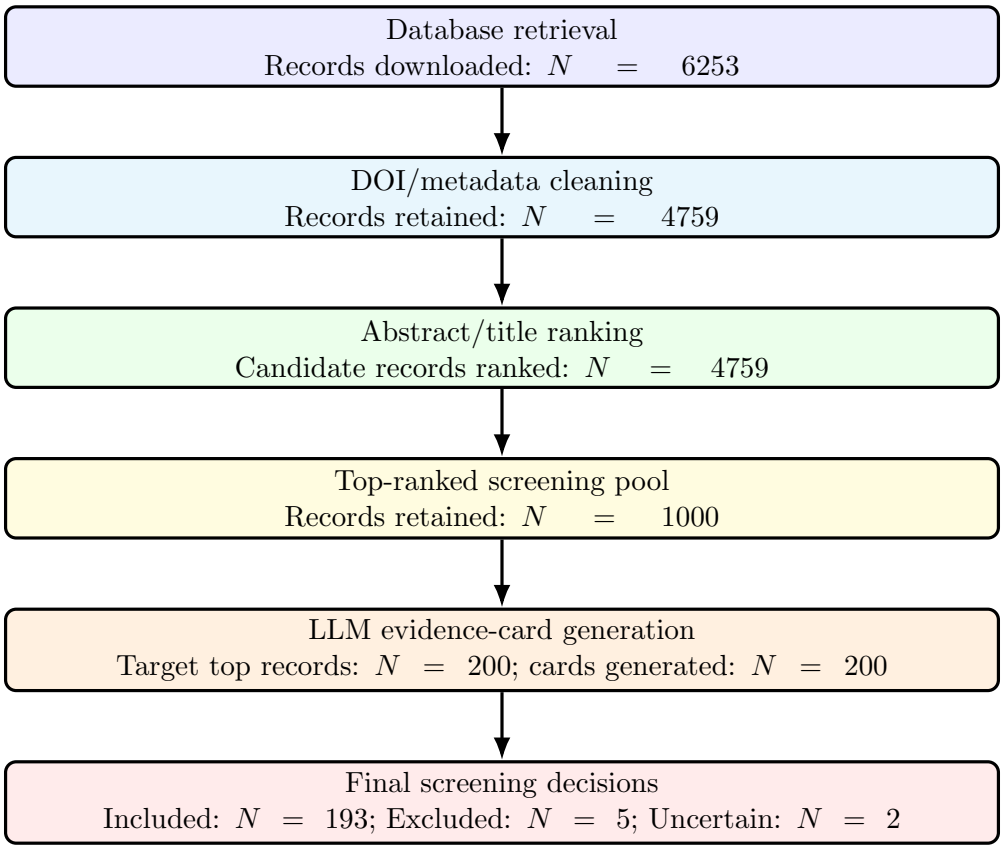


Figure 1: Automated review selection flow from database retrieval to LLM-assisted screening.

Table 1: Automated search, ranking, and screening summary.

Review stage	What was counted	Count	Source
Database retrieval	Records downloaded from configured bibliographic databases	6253	Search/articles.csv
DOI/metadata cleaning	Records retained after DOI/metadata cleaning	4759	Search/articles_clean_doi_only.csv
Abstract/title ranking	Candidate records ranked using abstract/title metadata	4759	Rankings/final_abstract_rank.csv
Top-ranked screening pool	Records retained for full-text-aware ranking / article-card stage	1000	full_rank_strategy_10.csv
LLM card target	Top records targeted for LLM evidence-card generation	200	TOP_N
LLM cards generated	Article cards available for synthesis	200	Step16 article_json
Cards used for writing	Cards selected for this article package	193	Step18.1 selected cards
Included	Cards with final screening decision = include	193	Step16 final_decision
Excluded	Cards with final screening decision = exclude	5	Step16 final_decision
Uncertain	Cards with final screening decision = uncertain	2	Step16 final_decision
Full text available	Cards processed with full text available	196	Step16 has_fulltext
Abstract/metadata only	Cards processed without full text	4	Step16 has_fulltext

Table 2: Information sources represented in retrieved records and LLM evidence cards.

Information source	Retrieved records	Evidence cards
MEDLINE	0	0
PubMed	1288	76
Scopus	0	0
PsycINFO	0	0
OpenAlex	1762	140
Crossref	1871	80
Semantic Scholar	1029	30
Europe PMC	2241	147

Table 3: Publication year distribution among selected evidence cards.

Publication year	Evidence cards
2020	184
2021	9

Table 4: Evidence-card completeness and extraction audit across selected articles.

Evidence item	Cards with item	Coverage
DOI present	193	100.0%
Full text available	189	97.9%
Article type extracted	193	100.0%
Aim/purpose extracted	193	100.0%
Domain/context extracted	193	100.0%
Methods/tools extracted	193	100.0%
Datasets/data sources extracted	190	98.4%
Outcomes/targets extracted	193	100.0%
Evaluation metrics extracted	140	72.5%
Main findings extracted	193	100.0%
Limitations/future work extracted	188	97.4%
Quality evidence extracted	165	85.5%

4 Main findings

4.1 Overview of canonical themes

Eight canonical themes emerged from the evidence-card completeness and extraction audit of the 193 included studies, with the understanding that individual studies may belong to more than one theme (Table 5). The themes span the roles of social media as a vector for misinformation, a sensor of public sentiment and mental health, a tool for disease surveillance, a channel for government communication and health promotion, a mirror of public concerns and risk perception, a site of stigma and conspiracy theories, and a domain shaped by cross-platform and cross-cultural dynamics.

Table 5: Summary of canonical themes, study counts, and key patterns.

Theme	**Studies**	**Key patterns**
Misinformation and the Infodemic	75	Social media as primary vector for rapid spread of false claims; fact-checking lags behind misinformation; shift to computational detection methods.

Table 5 (continued)

Theme	Studies	Key patterns
Public Sentiment and Mental Health	64	Consistent positive association between COVID-19 information exposure and negative mental health outcomes; one study found no association, indicating possible moderators.
Health Communication and Preventive Behaviors	38	Social media used for real-time health messaging but also amplifies misinformation; trust in government sources associated with better adherence.
Government Communication and Public Engagement	37	Low initial engagement; dialogic features enhance engagement; cross-country variation in outreach strategies.
Disease Surveillance and Prediction (Infoveillance)	27	Social media posts predict case counts up to 2 weeks ahead; lead times vary by region; evolution from correlation to causal inference methods.
Public Concerns and Risk Perception	24	Public attention evolves in stages; sentiment shifts from negative to neutral; misinformation triggers panic and distorted risk perception.
Cross-Platform and Cross-Cultural Dynamics	20	Weibo focuses on health; Twitter focuses on politics; false information more prevalent on Twitter; multilingual datasets enable temporal tracking.
Stigma and Conspiracy Theories	19	Rapid proliferation of stigmatising language and conspiracy theories; driven by ordinary citizens and verified users; computational tracking increasingly used.

4.2 Theme-level evidence patterns

Misinformation and the Infodemic. A robust cross-study pattern confirms that social media platforms functioned as primary vectors for the rapid dissemination of false and misleading health information during the pandemic. Misinformation propagated faster than accurate information, with health-related fake news dominating the landscape Naeem et al. [2020], Islam et al. [2020b], van der Linden et al. [2020]. The infodemic undermined public health behaviors and distorted risk perception, while fact-checking and psychological inoculation were proposed as countermeasures, though fact-checking often spread more slowly than the misinformation it sought to correct Tasnim et al. [2020], Shahi et al. [2021]. Methodological approaches evolved from manual content analysis to computational techniques such as social network analysis and natural language processing, enabling large-scale monitoring Ahmed et al. [2020b], Islam et al. [2020a].

Public Sentiment and Mental Health. Multiple studies leveraging sentiment analysis and topic modeling documented a significant correlation between COVID-19-related information exposure and heightened public panic, anxiety, and depression across diverse populations Ahmad and Murad [2020], Radwan et al. [2020], Yang et al. [2020]. Adverse emotions such as fear and anger mirrored pandemic case trajectories Medford et al. [2020], Zhao et al. [2020]. One notable contrast emerged from a Romanian study that found no direct association between COVID-19 information exposure and anxiety or depression, suggesting the presence of moderating factors such as emotion regulation strategies Cordoș and Bolboacă [2020]. Methodological sophistication increased from basic surveys to advanced machine learning models assessing emotion regulation strategies Yang

et al. [2020], Tang et al. [2020].

Health Communication and Preventive Behaviors. Social media enabled real-time public health communication but also facilitated the spread of misinformation. Exposure to misinformation on social media was consistently linked to greater misperceptions about COVID-19 and reduced adherence to preventive behaviors like social distancing Bridgman et al. [2020], Fridman et al. [2020]. Government sources were the most trusted and positively associated with accurate knowledge and adherence, whereas trust in private sources and social networks was negatively associated Fridman et al. [2020]. Misinformation proved resistant to simple correction, necessitating strategies addressing emotional, cognitive, and contextual factors Chou et al. [2020]. Co-designed social media campaigns demonstrated that non-traditional agents could achieve engagement levels comparable to specialized health media Graffigna et al. [2020], Pérez-Escoda et al. [2020].

Government Communication and Public Engagement. Government social media communication often faced low public engagement initially, with engagement patterns shifting as the pandemic evolved Liao et al. [2020], Hou et al. [2020], Pang et al. [2020]. Cross-country comparisons revealed significant variation: Singapore's Ministry of Health posted more frequently but attracted fewer comments per post than the U.S. CDC Sesagiri Raamkumar et al. [2020a,b]. Government posts emphasized instrumental support and official actions, while personal posts expressed more empathy and blame Liao et al. [2020]. Dialogic features and narrative styles enhanced engagement, whereas media richness sometimes reduced it Chen et al. [2020a], Ngai et al. [2020].

Disease Surveillance and Prediction (Infoveillance). Social media data consistently served as leading indicators of COVID-19 case counts, with several studies reporting statistically significant correlations and predictive power up to 14 days ahead of official statistics Shen et al. [2020], Hswen et al. [2020], Li et al. [2020b,a]. Lead times varied by region: from 3 days in China to 19 days in the USA using a digital decision-support tool Hswen et al. [2020]. Internet search data showed maximum correlations (r ¿ 0.89) with laboratory-confirmed cases at lags of 8–12 days Li et al. [2020a]. Methodological approaches progressed from simple cross-correlation analyses to Granger causality and machine learning classifiers, reflecting an emphasis on causal inference and temporal precision Shen et al. [2020], Li et al. [2020b]. Most studies, however, rely on correlational evidence because data cannot be linked to individual cases for privacy reasons, and models trained on Chinese samples may not generalize globally Hswen et al. [2020], Whitelaw et al. [2020].

Public Concerns and Risk Perception. Social media data revealed that public attention to COVID-19 evolved in distinct stages, with sentiment shifting from negative to neutral over time Zhao et al. [2020], Zhao and Xu [2020]. Misinformation triggered panic buying and distorted risk perception Hou et al. [2020], van der Linden et al. [2020]. Internet search and social media data often preceded official case reports by 8–14 days, suggesting potential for early warning systems Li et al. [2020a]. Contrasts emerged between studies that found delayed official information amplified negative emotions and those that noted timely clarifications reduced irrational behavior Hou et al. [2020], Liao et al. [2020].

Cross-Platform and Cross-Cultural Dynamics. Comparative studies revealed marked differences in discourse across platforms: Weibo users tended to emphasize health aspects, while Twitter users focused on policy and politics Chen et al. [2020b, 2021]. False information was more prevalent on Twitter than on Weibo, though science-based evidence also had higher presence on Twitter Pulido Rodríguez et al. [2020]. Multilingual Twitter datasets enabled tracking of temporal and geographic trends, with WHO announcements serving as major communication signals Pobiruchin et al. [2020]. Cross-country comparisons of public health authorities on Facebook showed varying outreach strategies and public sentiment, with Singapore's MOH receiving more favorable responses than the U.S. CDC or UK PHE Sesagiri Raamkumar et al. [2020a,b].

Stigma and Conspiracy Theories. Stigmatizing language and conspiracy theories proliferated

rapidly on social media. A near ten-fold increase in stigmatizing terms like 'Chinese virus' followed high-profile political references Budhwani and Sun [2020]. Conspiracy theories such as the 5G–COVID link and FilmYourHospital spread through ordinary citizens and verified users rather than automated bots Bruns et al. [2020], Ahmed et al. [2020a,b]. These narratives frequently employed group labeling and blame attribution while downplaying actual viral peril Li et al. [2020c]. Studies increasingly adopted computational methods—social network analysis, natural language processing, and automated content analysis—to track the spread in real time Budhwani and Sun [2020], Shahsavari et al. [2020].

4.3 Methods, tools, and data sources

Across the eight themes, the dominant data sources were Twitter and Sina Weibo, with Facebook, YouTube, and Instagram appearing less frequently. The most common methods were content analysis (both manual and automated), sentiment analysis, topic modeling (e.g., latent Dirichlet allocation), and social network analysis. Machine learning classifiers and natural language processing pipelines became increasingly standard for large-scale monitoring, especially within the misinformation and infoveillance themes. Several studies deployed the Linguistic Inquiry and Word Count program for psycholinguistic analysis Hyzer and Jerisha [2020]. Cross-platform comparisons required custom feature extraction frameworks, but universally designed workflows remained rare Chen et al. [2020b]. Surveys complemented computational analyses, particularly in the mental health and health communication themes. A notable gap was the limited use of qualitative follow-up methods (e.g., interviews, focus groups) to deepen understanding of quantitative patterns.

4.4 Outcomes, metrics, and evaluation practices

The most frequently reported outcomes were changes in sentiment polarity, volume of misinformation-related posts, engagement metrics (likes, shares, comments), and correlations between social media activity and official case counts. Studies evaluating government communication primarily used reach and engagement metrics, alongside sentiment polarity and trust-repair indices. In the infoveillance theme, metrics evolved from simple lag correlations to Granger causality tests and machine learning classifiers, reflecting a move toward causal inference. The mental health theme employed standardized scales (e.g., WHO-Five Well-Being Index, Generalised Anxiety Disorder Scale) alongside custom sentiment measures. A recurring evaluation trend was the shift from cross-sectional surveys to real-time, computationally intensive monitoring, though the lack of formal quality assessment tools beyond manual coding and expert annotation was a common limitation.

4.5 Cross-cutting limitations and open problems

Several limitations recur across the literature. Most studies relied on single-platform data (primarily Twitter or Weibo), limiting generalizability across platforms and languages. The predominance of Chinese-language samples and the early-pandemic time window further constrained external validity. Many key studies were preprints, raising questions about the robustness of conclusions. Few investigations linked social media sentiment or misinformation exposure directly to behavioral outcomes such as vaccination uptake or sustained social distancing, leaving a critical gap in understanding real-world impact. Additionally, the lack of universally designed content analysis frameworks hindered cross-platform comparisons, and the persistent challenge of scaling fact-checking and verification efforts remained unresolved. Future research should integrate behavioral and social science insights, employ longitudinal and multi-platform designs, and develop standardized metrics for tracking both the spread and the effects of online health misinformation.

5 Discussion

5.1 Interpretation of the evidence

The eight themes identified through this synthesis collectively depict social media as a dual-edged force during the COVID-19 pandemic. On one hand, platforms served as powerful channels for real-time disease surveillance, health communication, and public engagement, enabling the rapid dissemination of official guidance and the tracking of public sentiment and concerns. On the other hand, the same networks amplified misinformation, stigmatizing language, and conspiracy theories, often undermining public health efforts and distorting risk perception. A consistent cross-study pattern is the temporal alignment between social media activity and epidemic milestones: spikes in discourse on platforms like Weibo and Twitter frequently preceded official case reports, suggesting that social media can function as an early-warning system for emerging outbreaks Shen et al. [2020], Li et al. [2020a]. However, the same data stream also carried false claims that propagated faster than corrections, and exposure to such misinformation was associated with reduced adherence to preventive behaviors and increased psychological distress Bridgman et al. [2020], Ahmad and Murad [2020].

Contrasts across platforms and cultural contexts further nuance the evidence. Comparative studies revealed that Chinese Weibo users tended to emphasise health-related content, while Twitter users focused more on policy and political dimensions Chen et al. [2021]. Similarly, false news circulated less on Weibo than on Twitter, but science-based evidence also appeared more frequently on the latter Pulido Rodríguez et al. [2020]. These findings indicate that the design, governance, and user base of each platform shape the nature of public discourse and the effectiveness of health communication strategies. Government communication efforts, while generally well-intentioned, often struggled to maintain public engagement over time; posts that relied solely on situation updates lost traction, whereas narratives that incorporated empathy and dialogue generated higher interaction Liao et al. [2020], Chen et al. [2020a]. This suggests that successful crisis communication requires adaptive, emotionally resonant messaging tailored to the evolving concerns of the audience.

The synthesis also highlights an important methodological progression: early manual content analyses have given way to sophisticated computational approaches—machine learning, natural language processing, social network analysis—that enable large-scale, near-real-time monitoring of social media data. Yet the evidence remains largely correlational, and few studies have linked online discourse to observable health behaviours or long-term outcomes.

5.2 Strengths of the evidence base

The evidence base for this scoping review is robust in several respects. A total of 193 studies were included after screening a top-ranked pool of 1,000 records, covering a wide range of research designs from qualitative content analyses to large-scale predictive modelling using machine learning. The included studies span multiple social media platforms (Twitter, Sina Weibo, Facebook, YouTube, Instagram) and diverse linguistic and cultural contexts, enabling cross-platform and cross-cultural comparisons. Recurring methodological tools such as sentiment analysis, topic modelling, and social network analysis have been applied consistently, allowing synthesis of patterns across studies. Moreover, the availability of large, publicly shared datasets (e.g., multilingual COVID-19 Twitter corpora) has facilitated replication and validation of key findings. The breadth of the evidence—encompassing public sentiment, misinformation dynamics, disease surveillance, government communication, and stigma—provides a multifaceted understanding of social media's role during the pandemic. The use of an evidence-card completeness and extraction audit further strengthens confidence in the synthesised findings, even in the absence of formal quality scores.

5.3 Gaps and limitations in the literature

Despite the volume of research, several important gaps persist. First, most studies rely on data from a single platform (predominantly Twitter or Sina Weibo), which limits generalisability to other social media ecosystems and user demographics. Second, the majority of analyses cover only the early phases of the pandemic (January–April 2020), leaving the long-term evolution of online discourse and its impact on behaviour largely unexplored. Third, while correlations between social media signals and epidemiological indicators are repeatedly reported, few studies establish causal links—privacy constraints prevent linking individual posts to confirmed cases or health outcomes, confining evidence to ecological associations. Fourth, the synthesis reveals a notable lack of integration between computational methods and behavioural or social science theories. For example, misinformation research has benefited from psychological inoculation frameworks, but many studies still treat misinformation as a purely informational problem without accounting for emotional, cognitive, and contextual moderators van der Linden et al. [2020]. Fifth, cross-cultural comparisons remain limited; many comparative studies are confined to English-language platforms or specific country pairs, obscuring the dynamics in regions with different media ecosystems. Finally, the evidence base underrepresents studies that directly measure the behavioural consequences of social media exposure—such as actual vaccination uptake or adherence to social distancing—rather than self-reported intentions or beliefs.

5.4 Implications for future research

Future research should address these gaps by adopting longitudinal, multi-platform designs that capture discourse across the full pandemic trajectory and beyond. The development of universally applicable content analysis and feature extraction frameworks would enable more meaningful cross-platform and cross-cultural comparisons Chen et al. [2021]. Methodologically, there is a pressing need to move from correlational designs toward quasi-experimental or natural-experimental approaches that can identify causal effects of social media exposure on health behaviours, potentially leveraging platform-level interventions or policy changes as natural experiments. Integrating insights from behavioural science—such as the role of emotions, trust, and social norms—into computational models will enhance the understanding of why misinformation spreads and how it can be effectively countered. Additionally, future work should expand beyond convenience samples from a single platform to include underrepresented populations and languages, as well as assess the long-term mental health and behavioural consequences of sustained exposure to both accurate and misleading COVID-19 information. Finally, government and public health agencies should consider co-designing social media strategies with citizens, employing dialogic and empathic communication styles that have shown promise in maintaining engagement and trust Chen et al. [2020a], Liao et al. [2020].

5.5 Limitations of the automated review process

This scoping review was conducted using an automated screening and evidence extraction pipeline, which introduces specific limitations. From an initial download of 6,253 records, 4,759 remained after DOI metadata cleaning. The top 1,000 records ranked by relevance were selected for screening, and the LLM screened the top 200, generating 200 evidence cards. Of these, 193 were included in the final synthesis (5 excluded, 2 uncertain), and 196 of the 200 cards were based on full-text access. The reliance on a top-ranked screening pool means that some relevant studies falling below the rank threshold may have been missed, introducing selection bias toward studies with higher initial relevance scores. Furthermore, the LLM-based extraction of evidence cards may have introduced errors or omissions, although an evidence-card completeness and extraction audit was performed to

mitigate this risk. The automated process also precluded the use of formal quality assessment tools; instead, we assessed completeness and extraction consistency. Additionally, the review was limited to articles published in English or Chinese (based on the search strategy), potentially excluding relevant findings in other languages. Despite these constraints, the approach enabled a systematic and reproducible synthesis of a large body of literature, providing a comprehensive mapping of the evidence.

6 Conclusion

6.1 Overall conclusion

This scoping review synthesises evidence on the multiple roles social media played during the COVID-19 pandemic, drawn from eight thematic clusters that collectively reflect the platform's duality. The largest bodies of work—on *Misinformation and the Infodemic* (75 studies) and *Public Sentiment and Mental Health* (64 studies)—reveal that social media simultaneously served as a vector for rapidly propagating false claims and as a mirror of pandemic-driven emotional distress. Other major themes, including *Health Communication and Preventive Behaviors* (38 studies), *Government Communication and Public Engagement* (37 studies), and *Disease Surveillance and Prediction* (27 studies), illustrate its potential as a tool for public health messaging, official engagement, and near-real-time outbreak monitoring. Smaller but distinct clusters—*Public Concerns and Risk Perception* (24 studies), *Cross-Platform and Cross-Cultural Dynamics* (20 studies), and *Stigma and Conspiracy Theories* (19 studies)—further highlight the context-dependent nature of social media discourse. Because studies could belong to more than one theme, the theme counts sum to more than the total number of included records. Across all themes, social media emerges as a double-edged instrument: it amplifies both health-promoting information and harmful narratives, and its effects are moderated by platform affordances, trust in sources, and cultural context.

6.2 Most important evidence patterns

Several cross-cutting patterns emerge from the synthesis. First, misinformation consistently propagates faster than accurate information on social media, with health-related false claims dominating discourse and proving resistant to correction efforts such as fact-checking Naeem et al. [2020], Islam et al. [2020b], van der Linden et al. [2020]. Second, exposure to COVID-19 content on social media is repeatedly associated with heightened anxiety, depression, and panic, although one study diverged from this pattern, suggesting moderating factors exist Ahmad and Murad [2020], Radwan et al. [2020], Cordoș and Bolboacă [2020]. Third, social media data—especially from platforms such as Weibo and Twitter—demonstrates predictive value for disease surveillance, with lead times of up to two weeks before official case counts, using methods such as Granger causality and lag correlation Shen et al. [2020], Hswen et al. [2020], Li et al. [2020b]. Fourth, government communication on social media initially faced low public engagement, with patterns shifting as the pandemic evolved; dialogic and empathic messaging outperformed purely informational posts Liao et al. [2020], Sesagiri Raamkumar et al. [2020a], Chen et al. [2020a]. Fifth, cross-platform comparisons reveal marked differences in discourse: Weibo users emphasized health aspects, while Twitter users focused on policy and politics, and false information was more prevalent on Twitter than on Weibo Chen et al. [2020b], Pulido Rodríguez et al. [2020]. Finally, stigmatizing language and conspiracy theories spread rapidly, often fueled by verified accounts rather than bots, and computational methods increasingly track these dynamics in real time Budhwani and Sun [2020], Ahmed et al. [2020a], Li et al. [2020c].

6.3 Practical and research implications

For public health practitioners, the findings underscore the need for proactive, empathic, and dialogic communication strategies that build trust and engage the public rather than merely disseminating information. Real-time monitoring of social media for misinformation and sentiment shifts should be integrated into outbreak response frameworks, while fact-checking efforts must be complemented by interventions that address emotional, cognitive, and contextual drivers of belief in false claims van der Linden et al. [2020], Tasnim et al. [2020]. For researchers, the review highlights persistent methodological limitations, including reliance on single-platform samples, short observation windows, and ecological correlations that cannot establish causality. Future research should adopt cross-platform and longitudinal designs, integrate behavioral and social science theories into misinformation studies, and develop standardised content-analysis workflows to enable meaningful comparisons across cultures and languages Whitelaw et al. [2020], Hyzer and Jerisha [2020]. The predominance of Chinese-language and English-language datasets also calls for greater linguistic and geographic diversity in sampling.

6.4 Future work

Several evidence gaps remain. Most studies analysed only one platform (e.g., Twitter, Weibo) over limited time periods, constraining generalisability across pandemic phases and sociocultural contexts Hswen et al. [2020], Shahi et al. [2021]. Future work should employ multi-platform, multilingual, and multi-country designs to capture how platform design and cultural norms shape discourse. Causal inference is needed to move beyond correlations between social media activity and health outcomes; natural experiments or quasi-experimental designs could help establish whether social media use directly influences mental health, vaccine uptake, or adherence to preventive measures. Longitudinal research should track the evolution of public sentiment, trust, and misinformation persistence over the full pandemic course, including post-acute phases. Qualitative methods—such as interviews or focus groups—remain underutilised and could reveal the lived experiences behind quantitative patterns Cordoș and Bolboacă [2020], Ölcer et al. [2020]. Additionally, the ethical implications of using social media data for surveillance and prediction, including privacy, consent, and algorithmic bias, require systematic examination. Finally, interdisciplinary collaboration between computer science, public health, and social science is essential to design scalable interventions that harness the positive potential of social media while mitigating its harms.

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